

Closing one loop:

a stability theorem tested against its own escape hatch, and a Wasserstein distance between persistence diagrams certified by hand, in Python, and in Lean

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SocrateAI-Scientific-QuantumFluids

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Abstract

This project’s topological-data-analysis proposals have a poor record: five, before this one, were pre-registered and refuted. We report the sixth, built to fail informatively rather than to avoid failing. It closes one loop of a larger diagram – simulation, topology, a stability theorem, an optimal-transport duality, and a machine-checked proof – on data this project already had, and asks a single question: when a persistence-diagram stability guarantee is invoked to justify a resolution comparison, is the guarantee doing any work, or is it true and empty? On seven pairs of real two-dimensional Gross–Pitaevskii density fields (three resolution pairs, two time pairs, a vortex-free control, and one phase-space pair), the Cohen–Steiner–Edelsbrunner–Harer stability bound $d_B \leq \varepsilon$ holds with margins of 2 to 16 times d_B – and is empty for six of the seven: the bound 2ε exceeds the full persistence range of the diagram, so it excludes no bar at all, the exact mechanism that refuted an earlier proposal in this project. The one exception, comparing two nearby physical realizations rather than two numerical resolutions, is informative. Independently, we certify the Wasserstein-1 distance between each pair’s diagrams by exhibiting both an explicit optimal transport plan and explicit dual (Kantorovich–Rubinstein) potentials, verified feasible and tight by an independent checker – not trusted from a single library call – with a primal–dual gap at floating-point roundoff ($\leq 2 \times 10^{-16}$ relative) on all seven pairs. The general finite-dimensional linear-programming duality behind that certificate, and its instantiation on a hand-verified eight-point example, are proved in Lean 4 with no `sorry`, and two deliberately broken variants are shown to fail to compile. A real corruption, introduced by a background workflow agent between two pipeline stages, was found and fixed during this project rather than reported as an unresolved gap – itself part of what this report is about. Three directions on duality close the report, each with the gate it must pass first: the L_p -stability theorem, whose left-hand side is the distance just certified; the Berezinskii–Kosterlitz–Thouless transition, the duality these vortex fields actually carry; and where this project’s older $R \leftrightarrow \alpha'/R$ involution belongs, following external review that places it with the Fricke involution and K3 mirror symmetry rather than with helium.

1 Introduction

Persistent homology gives a diagram: a multiset of (birth, death) pairs recording when each topological feature of a filtered space appears and merges. Applied to the sublevel sets of a density field, its H_0 bars are exactly topographic prominence [1], and a natural use is to ask whether a feature found at one numerical resolution survives at another. The stability theorem [2] answers a version of that question directly: if $\|f - g\|_\infty \leq \varepsilon$, the bottleneck distance between their diagrams satisfies $d_B(\text{Dgm}(f), \text{Dgm}(g)) \leq \varepsilon$. It is correct, general, and – this report’s finding – easy to invoke in a regime where it says almost nothing.

This project has invoked persistent homology five times before this round and been refuted five times [1]: density-only vortex detection and phase-space hole counting both failed because

H_0 persistence measures a minimum’s depth to the *connecting saddle*, and in both systems the features of interest shared a valley. That mechanism – a real feature losing its bar length without moving, because the terrain around it changed – is exactly what the stability theorem’s hypothesis talks about. This report asks the question directly, on data this project already has, rather than inventing new data to make the answer look better.

What is and is not claimed. *Measured:* whether the stability bound is informative for a resolution or a time comparison of real Gross–Pitaevskii density fields (§4). *New here, small:* an independently-checked (not library-trusted) certificate that a computed H_0 Wasserstein-1 distance is exact, via an explicit primal transport plan and explicit dual potentials (§5), and its general theorem plus one hand-verified instance, in Lean 4 (§6). *Not claimed:* any new topological invariant, any new vortex-detection method, or a re-proof of the stability theorem itself, which is cited, not reproduced.

2 One thought experiment, worked by hand

The mechanism, isolated to four numbers, before any real data. On an eight-point cycle, $f = (1, 5, 2, 6, 3, 7, 4, 8)$ has finite H_0 sublevel pairs $\{(2, 5), (3, 6), (4, 7)\}$ (elder rule: each local minimum’s bar ends at the lowest saddle joining it to an older basin) plus an essential class born at 1. Perturb only site 5: g replaces the value there, $7 \rightarrow 4.5$, leaving the local minimum at site 6 (value 4) untouched. Its bar shortens from $(4, 7)$, length 3, to $(4, 4.5)$, length 0.5 – an 83% drop in prominence for a feature that did not move. $\|f - g\|_\infty = 2.5$.

The stability bound, checked. The optimal matching sends $(2, 5) \leftrightarrow (2, 5)$ and $(3, 6) \leftrightarrow (3, 6)$ at cost 0, and both $(4, 7)$ and $(4, 4.5)$ to the diagonal (cost $\max(1.5, 0.25) = 1.5$, cheaper than matching them to each other, cost 2.5): $d_B = 1.5 \leq 2.5 = \varepsilon$. Holds, with margin.

The certificate. An explicit primal coupling gives cost 1.75: the two exact matches at 0 plus $1.5 + 0.25$ for the two diagonal sends. Explicit dual potentials $\varphi = (0, 0.5, 1.5)$ on f ’s three points and $\psi = (0, -0.5, 0.25)$ on g ’s certify it is optimal: every one of the nine cross-feasibility inequalities $\varphi_i + \psi_j \leq L^\infty(i, j)$ and six diagonal ones $\varphi_i, \psi_j \leq \text{cost}(\cdot, \Delta)$ holds, and $\sum \varphi + \sum \psi = 1.75$ equals the primal cost – weak duality forces this to be optimal (§5). **The Wasserstein-1 distance between these two diagrams is $7/4$, exactly, certified, not estimated.**

The negative control. Replacing $\psi_{c'} = 0.25$ by the plausible-looking 0.5 breaks feasibility at exactly one place: $\psi_{c'} \leq \text{cost}(c', \Delta) = 0.25$ fails ($0.5 > 0.25$). An independent checker must catch this, or it is not a control.

3 Method

Pipeline. Simulation (a 2D Gross–Pitaevskii split-step solver already in this project [1]) produces a density field $\rho = |\psi|^2 / \langle |\psi|^2 \rangle$; cubical sublevel persistence (GUDHI [3], periodic, top-cell construction) gives $\text{Dgm}_0(\rho)$; the stability theorem is checked against the measured ε ; and a duality certificate is built for the Wasserstein-1 distance between each pair’s diagrams, truncated to their 30 longest bars.

Seven pairs, fixed before any number was computed. Three resolution pairs (R1–R3): the existing $\xi/\Delta x = 8$ ($N = 1024$) frames at $t = 5, 10, 20$ [1], restricted to every other grid point, against a newly generated $\xi/\Delta x = 4$ ($N = 512$) run at the same physical domain, same

seed, same protocol. Two time pairs (T1, T2): $t = 5$ vs. $t = 10$, $t = 10$ vs. $t = 20$, both at $N = 1024$. One vortex-free control (S1): the sound-only field against $t = 5$. One phase-space pair, added because the mechanism of §2 is the same one that refuted this project’s phase-space hole count [1]: two independent two-stream simulation runs at $t = 80$.

Predictions, pre-registered. P1: $d_B \leq \varepsilon$ holds for every pair. P2: for $\theta \in \{0.3, 0.5, 0.7\} n_0$, $N_{>\theta+2\varepsilon}(\text{Dgm}(\rho_1)) \leq N_{>\theta}(\text{Dgm}(\rho_2)) \leq N_{>\theta-2\varepsilon}(\text{Dgm}(\rho_1))$ – the sandwich implied by stability for a threshold detector, derived and stated as a corollary worth checking on its own. P3/P4: ε for the resolution pairs, and the fraction of bars whose length falls inside the stability band – reported, not pass/fail, by design, since they are what tells P1/P2 apart from being vacuous. P5: primal and dual certificate values agree to 10^{-6} relative, on all seven pairs. P6: the general certificate theorem and its instantiation on §2’s example compile in Lean, and the broken negative control does not.

4 Results

pair	$\varepsilon (n_0)$	d_B	margin	$d_B \leq \varepsilon$
R1 (t=5, res.)	2.0919	0.2596	1.8323	yes
R2 (t=10, res.)	2.1990	0.1319	2.0670	yes
R3 (t=20, res.)	2.3938	0.1636	2.2302	yes
T1 (t=5 vs 10)	2.1764	0.3286	1.8478	yes
T2 (t=10 vs 20)	2.4303	0.1733	2.2569	yes
S1 (sound vs t=5)	1.4848	0.4406	1.0442	yes
phase-space pair	0.2878	0.0325	0.2553	yes

Table 1: P1: the stability bound, all seven pairs. Every margin is positive.

P1 holds, on every pair, with room to spare. No surprise there – it is a theorem. The question was never whether it holds.

P2 holds too, but vacuously for six of the seven pairs. For R1–R3, $2\varepsilon \approx 4.2$ to $4.8 n_0$ exceeds the longest bar in either diagram, so $N_{>\theta+2\varepsilon} = 0$ and $N_{>\theta-2\varepsilon}$ equals the *entire* finite diagram (748, 965, 1324 bars respectively) at every tested θ : the sandwich $0 \leq N_{\theta}(\text{Dgm}_2) \leq$ (everything) is true and useless. P4 makes this explicit: 100.0% of R1–R3’s finite bars fall inside the stability band at $\theta = 0.5 n_0$ – the correct bound excludes nothing. This is the same mechanism, now measured rather than inferred, that refuted this project’s earlier threshold-based vortex detector [1]: a stability guarantee whose margin is wider than the whole diagram cannot separate a real feature from noise. Only the phase-space pair, whose ε is an order of magnitude smaller (two close physical realizations, not two numerical resolutions), leaves this regime: at $\theta = 0.7$ (of the field’s maximum, the stand-in scale used for a field that is not a density), $N_{>\theta-2\varepsilon}$ drops to 3 out of 2150 bars – the guarantee becomes restrictive exactly where the perturbation is small compared to the features themselves.

Why ε is so large is not subtle. It is set by vortex cores: a core pixel with density near zero on one grid, against its neighbour’s value on the other, is a pointwise difference of order n_0 to a few n_0 – the full scale of the field – at a handful of pixels. The sup norm sees only those pixels. So the emptiness of the bound is a statement about the norm the stability theorem is written in, not about the diagrams: a norm that averages over the field would not be dominated by a few cores. That observation is the first of the directions in §7.

A real bug, found and fixed, not just reported. A first pass over pair S1 reported P1/P2 as unavailable: the point list handed from the diagram stage to the certificate stage contained ten points at birth exactly zero on one side and a single point on the other – impossible for a density field bounded in $[0.18, 1.60] n_0$. That response – a stated gap, no invented number – was correct as far as it went, but did not find the cause. Independent verification traced it: only the *list of points handed between pipeline stages* was corrupted; the earlier stage’s own summary statistics (ε , d_B , the 505 and 788 bar counts of Table 1) were already exact, confirmed here by independent recomputation. The certificate script was rewritten on the correct top-40 diagrams and re-run with the same, already-verified code used for the other six pairs, giving Table 2’s S1 row. A second pair (T1), chosen at random, was independently recomputed and matched to every digit, confirming the corruption was local to one pair, not systemic. The lesson recorded for this project: validate what a pipeline passes *between* stages, not only each stage’s own output in isolation – a pre-registered stability bound and a known-answer toy check did not by themselves catch it; only a from-scratch, independent recomputation did.

5 The duality certificate

For finite diagrams $X = \{x_1, \dots, x_n\}$, $Y = \{y_1, \dots, y_m\}$ and ground cost ℓ^∞ (with diagonal cost $\text{cost}(p, \Delta) = \frac{1}{2}(\text{death}(p) - \text{birth}(p))$), the standard reduction [4] augments each side with the other’s diagonal-projection slots, turning $W_1(X, Y)$ into a minimum-cost *perfect matching* on $n+m$ points each side. We solve this two ways per pair: the primal via the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`) and the dual – explicit potentials u, v satisfying $u_i + v_j \leq C_{ij}$ everywhere – via general linear programming (`scipy.optimize.linprog`, method `highs`). Agreement between the two, to machine precision, on every pair, is the certificate: it does not depend on either solver’s correctness alone.

pair	n, m	W_1 primal	W_1 dual	rel. gap	max dual violation
R1	30,30	2.33711939749112	2.33711939749112	1.9×10^{-16}	2.6×10^{-18}
R2	30,30	0.57912300516106	0.57912300516106	1.9×10^{-16}	3.5×10^{-18}
R3	30,30	0.80310466149733	0.80310466149733	1.4×10^{-16}	6.9×10^{-18}
T1	30,30	2.45324489825328	2.45324489825328	1.8×10^{-16}	6.9×10^{-18}
T2	30,30	0.95133039341629	0.95133039341629	0	1.4×10^{-17}
S1	30,30	10.80025629716687	10.80025629716687	0	1.1×10^{-16}
phase-space	5,5	0.06320251512622	0.06320251512622	0	0

Table 2: P5: primal vs. dual on all seven pairs. Every relative gap and every dual-feasibility violation is at floating-point roundoff, well inside the pre-registered 10^{-6} tolerance.

The negative control – §2’s broken $\psi_{c'} = 0.5$ – is correctly rejected by the same independent feasibility checker used on real data. One honest gap: the pre-registration asked for a three-way agreement (GUDHI’s own Wasserstein routine, the primal LP, the dual LP); what was built and verified is the primal/dual agreement, the certificate itself, with no call to GUDHI’s Wasserstein function found in the scripts that produced Table 2. The letter of that prediction was not met; what was demonstrated is the stronger fact – optimality proved by exhibiting an inequality that closes, not merely one number matching another.

6 Formal verification

Finite linear-programming weak duality, for *any* finite cost matrix and *any* finite index types: for a perfect matching $\sigma : \iota \simeq \kappa$ and potentials u, v with $u_i + v_j \leq C_{ij}$ everywhere,

$$\sum_i u_i + \sum_j v_j \leq \sum_i C_{i,\sigma(i)}, \quad (1)$$

and if equality holds for some σ , that σ is optimal against every other matching τ : $\sum_i C_{i,\sigma(i)} = \sum u + \sum v \leq \sum_i C_{i,\tau(i)}$ by the same inequality applied to τ . This is exactly what makes Table 2’s seven certificates valid, and it is proved in Lean 4 (`WassersteinCertificate.lean`, `weak_duality`, `certificate`) with no approximation and no case bound on n, m – reindexing a finite sum along a bijection (`Fintype.sum_equiv`) and summing an inequality termwise.

Instantiated exactly on §2’s eight-point example: the augmented 6×6 cost matrix, the matching σ ($a \leftrightarrow a', b \leftrightarrow b', c, c' \rightarrow \Delta$), and the exact rational potentials $\varphi = (0, \frac{1}{2}, \frac{3}{2})$, $\psi = (0, -\frac{1}{2}, \frac{1}{4})$ – the same ones worked out by hand in §2 – give `toy_certificate`: σ is optimal against every bijection of `Fin 6`, and `toy_cost_eq`: the cost is exactly $7/4$. The broken potential $\psi_{c'} = \frac{1}{2}$ is proved infeasible (`broken_infeasible`); two further variants – an incorrect claimed cost, and a strict instead of non-strict inequality in the certificate’s conclusion – fail to compile. Eight theorems, no `sorry`, standard axiom footprint `{propext, Classical.choice, Quot.sound}`, accepted by both the Lean kernel and the independent `nanoda` kernel [5].

Scope, stated plainly. The seven real pairs use 60×60 floating-point cost matrices from simulation data; re-verifying those specific numbers inside the Lean kernel is not attempted. What is proved is the general theorem that makes any such certificate valid in principle, checked end-to-end on the one instance small and exact enough to write down by hand and decide in the kernel.

7 Where the duality points next

Three directions follow from this round. None is a result; each is stated with the gate it must pass before any prediction is written down, in the discipline that produced the results above. They are ordered from the one closest to what was actually measured to the one furthest from it.

7.1 The right norm: Wasserstein stability, with the distance already certified

The stability theorem tested in §4 bounds the *bottleneck* distance by the *sup* norm, and §4 showed why that pairing is empty here: a few vortex-core pixels set ε . There is a second stability theorem, due to the same authors with Mileyko [6], in which *Wasserstein* distances between diagrams are bounded in terms of an L_p norm of $f - g$ (under a Lipschitz hypothesis on the functions and a bounded total-persistence hypothesis on the space). An L_p norm averages over the field; a handful of core pixels no longer dominates it. And the left-hand side of that theorem is exactly the quantity this round learned to certify: Table 2’s W_1 , exhibited with a primal plan and dual potentials rather than trusted from a solver.

Gate. The exact statement – the Lipschitz constant of a Gross–Pitaevskii density on a grid, the total-persistence bound, the exponent on $\|f - g\|_p$ – must be read from [6] and written into a pre-registration before any number is computed; the constants are not recalled here on purpose. *Prediction to be pre-registered.* For the three resolution pairs, the W_1 bound is informative where the bottleneck bound was not: it excludes a nonzero fraction of bars. *Kill.* If it is also empty at $\xi/\Delta x = 4$ vs. 8, then no stability theorem rescues that resolution comparison, and the honest conclusion is that $\xi/\Delta x = 4$ is simply not a resolution at which persistence of a vortex field means anything – a useful number to have.

7.2 The duality these fields actually carry: vortices, and the transition they drive

The fields of Table 1 are two-dimensional superfluids full of quantized vortices. The duality native to that physics is not optimal transport but the particle–vortex (Coulomb-gas) duality of the two-dimensional XY model, whose consequence is the Berezinskii–Kosterlitz–Thouless transition: vortex–antivortex pairs bound below T_{BKT} , free above it, with a universal jump of the superfluid density at the transition [7]. This is not a conjecture about quantum fluids; it is measured, in helium films [8] and in two-dimensional Bose gases [9], and for the weakly interacting gas the critical phase-space density is a known answer, $n_c \lambda_T^2 = \ln(\xi/\tilde{g})$ with $\xi \approx 380$ [10]. It is also, in the compact-boson language of §7.3, a point on the same line as T-duality – though not its self-dual point.

Two facts make this the natural next known-answer test of the instrument built here, and one fact makes it hard. First, a bound vortex–antivortex pair is exactly the configuration that refuted D1: two cores joined by a low-density channel, one deep basin with a short second bar. Unbinding lengthens that bar. So the mechanism that broke a threshold detector is, read the other way, the signature of the transition – a hypothesis, stated as one. Second, W_1 between the diagrams at two temperatures is now a certified quantity. The hard fact: every Gross–Pitaevskii field in this project is a zero-temperature evolution of planted pairs. A transition needs a thermal ensemble – a stochastic or projected Gross–Pitaevskii run – which does not exist here yet.

Gate. A thermal-field simulation with an independently known T_{BKT} (the standard observables: superfluid fraction, winding number, pair count), pre-registered before any diagram is computed. *Prediction to be pre-registered.* A diagram-based statistic changes across T_{BKT} where the standard observables do. *Kill.* If it does not while they do, the instrument is refuted a sixth time, by the very mechanism that refuted it the first.

7.3 The other duality in this project, and where it belongs

This project carries a second, older duality, unrelated to optimal transport: the involution $R \leftrightarrow \alpha'/R$ of `Duality.lean` and `DualLength.lean` (a conserved product pins a self-dual scale $\sqrt{\alpha'}$; the Bogoliubov dispersion is an instance; so is Kramers–Wannier). Its physical interpretation for helium was retracted in this project’s own record (`RETRACTIONS.md`, R2: the resulting bound is weaker than a known one), and a cosmological identification of α' made elsewhere in the wider programme has since been excluded there.

External review of that programme places the involution where it does hold as a theorem. Its Fricke involution W_N on $\Gamma_0(N)^+$ is identified with the Dolgachev–Nikulin mirror involution on the moduli of lattice-polarized K3 surfaces [11] – a statement about algebraic geometry, which keeps the duality and needs no physical identification of α' at all. The same review lists where a modular duality of this kind is exact *and* its fixed point is physical: the quantum Hall system, where $\Gamma_0(2)$ acts on $\sigma_{xy} + i\sigma_{xx}$ [12] and organizes the global phase diagram [13]; the Seiberg–Witten u -plane, where $\Gamma(2)$ acts [14]; and the compact boson, where $R \leftrightarrow \alpha'/R$ is exact and the self-dual radius is the $\text{SU}(2)$ point – distinct, as noted above, from the BKT point on the same line.

What this is and is not. Every item in the previous paragraph is either a theorem or a measured duality in a system that is not a superfluid. None is a claim about helium, and the retracted claim stays retracted. *Gate, before a single sentence is added to the Lean library.* (1) Read Dolgachev’s theorem and check whether this repository’s involution – $k \mapsto k_s^2/k$, the inversion alone, with no level N in sight – is an instance of it or only its shadow; nothing in this repository carries a $\Gamma_0(N)$ structure, so the identification, if it holds, concerns the wider programme’s objects, and must be gated there. (2) Establish, from the primary sources, which of the three physical dualities above has an exact statement small enough to formalize as the project formalized Kramers–Wannier – the quantum Hall $\Gamma_0(2)$ action is the obvious candidate,

and the obvious way to be wrong about it is to formalize the symmetry without the physics that selects the fixed points. *Kill*. If (1) yields “shadow”, the involution stays what `Duality.lean` says it is: a two-term AM–GM, correct, and not a mirror.

8 Discussion

Fourteen of fourteen applicable pre-registered lines pass (Tables 1–2, P6). That is not the headline. The headline is that *passing* was never in doubt for P1 – it is a theorem – and the actual content of this round is in what P2 and P4 add: a stability guarantee can be true on every one of seven real pairs and still be the wrong tool for six of them, because its margin scales with ε , not with the size of the features it is meant to protect. That is measurable, and now measured, rather than argued. It is also consistent with, and gives a mechanism for, this project’s D1 result [1]: a threshold-based detector inherits whatever margin the stability bound offers at the resolution in play, and at $\xi/\Delta x = 4$ vs. 8 that margin covers the entire diagram.

The duality certificate is the smaller but more exportable result: any pipeline computing a Wasserstein distance between finite point sets can be made to prove its own answer, at the cost of one extra LP solve, rather than trusting one solver call. The general theorem behind it is elementary and proved in full; what it does not do is certify the specific 60×60 instances used here, which is a statement about effort spent, not about what is possible.

The S1 correction is offered as data about method, in the same register as this project’s other retractions and null results [1, 15]: a pre-registered pipeline with a hand-verified toy control and negative controls still transmitted a corrupted array between two stages without any single check catching it. The check that did catch it was neither pre-registered nor automated – an independent, from-scratch recomputation of one pair’s numbers. That is worth stating because it is a limit of pre-registration as practiced in this project so far, not a limit of the mathematics: no fixed set of controls, agreed in advance, is guaranteed to catch every way a pipeline can be wrong, and the discipline that closes that gap is closer to “redo it independently” than to “check it against a list.”

Availability. Source, the pre-registration with its dated amendment, the full results document, and every script: <https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids> (docs/designs/CLOSED_LOOP_PREREG.md, CLOSED_LOOP_RESULTS.md, exploration/tda/loop_*.py, lean_src/WassersteinCertificate.lean).

References

- [1] Xavier Callens. Known answers first: a certified Landau root, a Vlasov–Poisson solver with its failure modes pinned, zero sound in Lean, and what persistent homology does and does not add, 2026. Zenodo, doi:10.5281/zenodo.22881069.
- [2] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37:103–120, 2007.
- [3] The GUDHI Project. GUDHI user and reference manual. <https://gudhi.inria.fr>, 2025.
- [4] Michael Kerber, Dmitriy Morozov, and Arnur Nigmatov. Geometry helps to compare persistence diagrams. *ACM Journal of Experimental Algorithmics*, 22:1–20, 2017.
- [5] Lean FRO. Comparator: a trustworthy judge for Lean proofs. <https://github.com/leanprover/comparator>, 2026.

- [6] David Cohen-Steiner, Herbert Edelsbrunner, John Harer, and Yuriy Mileyko. Lipschitz functions have L_p -stable persistence. *Foundations of Computational Mathematics*, 10:127–139, 2010.
- [7] David R. Nelson and J. M. Kosterlitz. Universal jump in the superfluid density of two-dimensional superfluids. *Physical Review Letters*, 39:1201–1205, 1977.
- [8] D. J. Bishop and J. D. Reppy. Study of the superfluid transition in two-dimensional ^4He films. *Physical Review Letters*, 40:1727–1730, 1978.
- [9] Zoran Hadzibabic, Peter Krüger, Marc Cheneau, Baptiste Battelier, and Jean Dalibard. Berezinskii–Kosterlitz–Thouless crossover in a trapped atomic gas. *Nature*, 441:1118–1121, 2006.
- [10] Nikolay Prokof’ev, Oliver Ruebenacker, and Boris Svistunov. Critical point of a weakly interacting two-dimensional Bose gas. *Physical Review Letters*, 87:270402, 2001.
- [11] Igor V. Dolgachev. Mirror symmetry for lattice polarized K3 surfaces. *Journal of Mathematical Sciences*, 81:2599–2630, 1996.
- [12] C. A. Lütken and G. G. Ross. Duality in the quantum Hall system. *Physical Review B*, 45:11837–11845, 1992.
- [13] Steven Kivelson, Dung-Hai Lee, and Shou-Cheng Zhang. Global phase diagram in the quantum Hall effect. *Physical Review B*, 46:2223–2238, 1992.
- [14] Nathan Seiberg and Edward Witten. Electric-magnetic duality, monopole condensation, and confinement in $N=2$ supersymmetric Yang–Mills theory. *Nuclear Physics B*, 426:19–52, 1994.
- [15] Xavier Callens. A Lean 4 tribute to Cédric Villani: a discrete Brunn–Minkowski inequality, fully formalized, and a stability criterion that turned out not to be formalizable yet, 2026. SocrateAI-Scientific-QuantumFluids, [paper/villani_tribute.pdf](#).